



Star-Galaxy Separation in EDFs

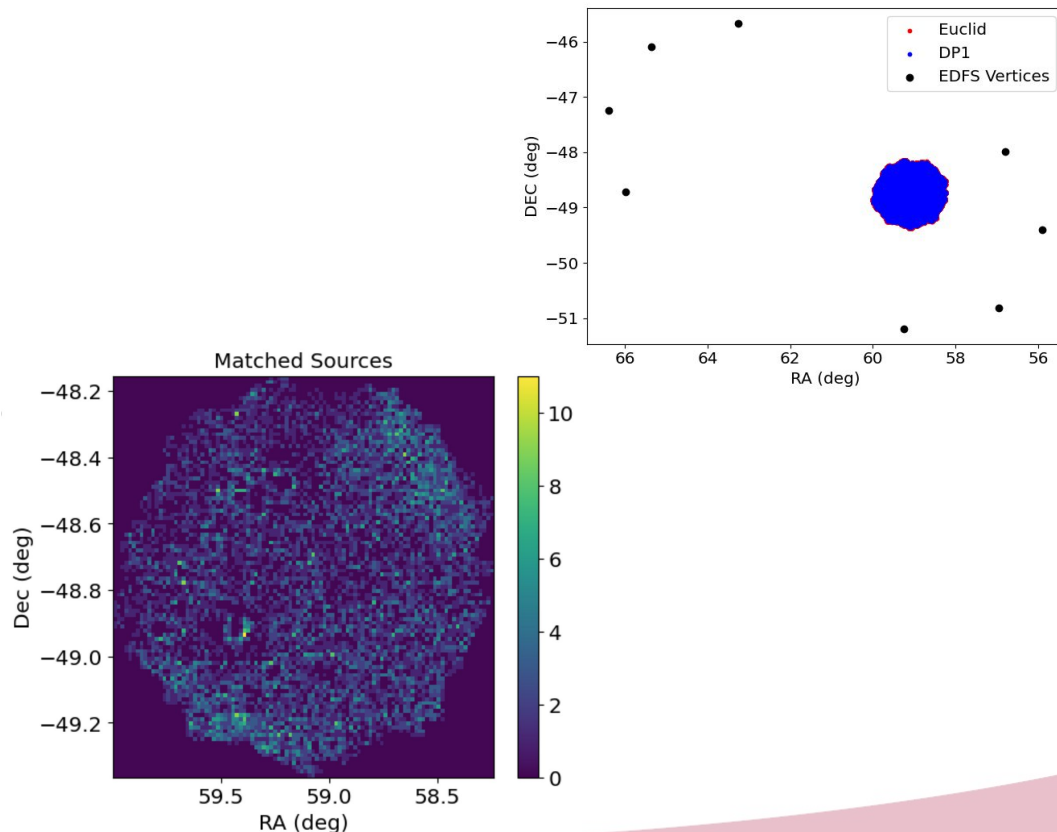
Preliminary photometry using LSSTComCam + Euclid

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Merged Rubin-Euclid EDFS Catalog

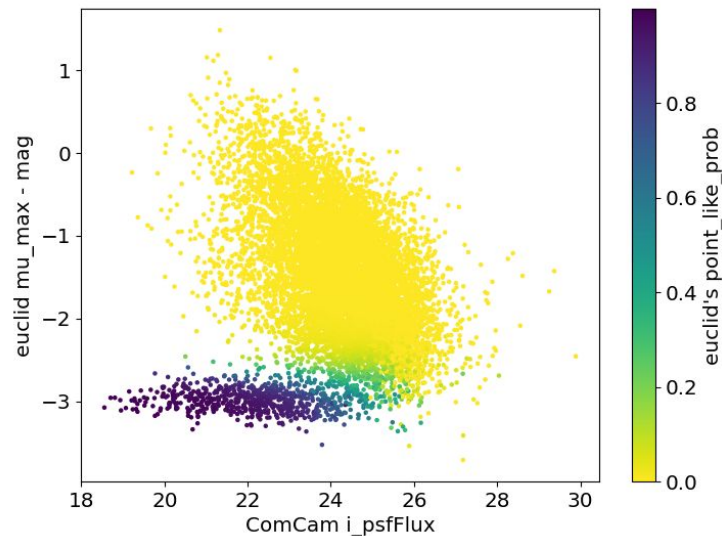
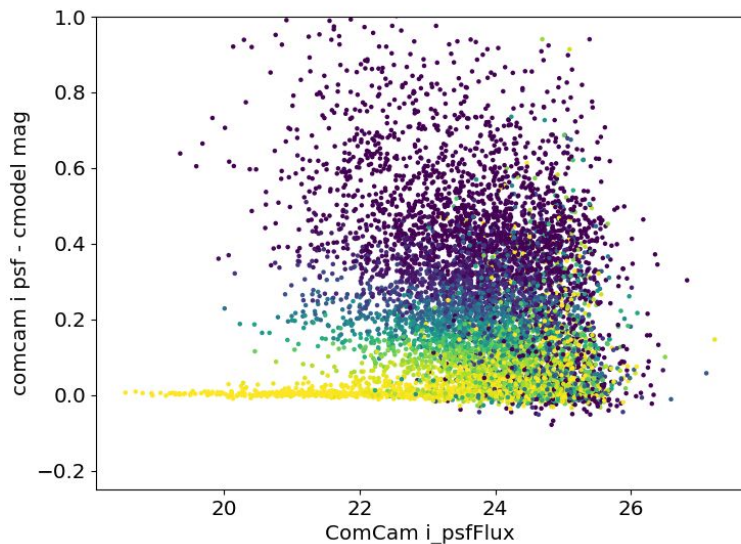
1. Peter Ferguson uploaded the Euclid data which overlapped DP1 to USDF
2. Queried DP1 coadd object catalog within Euclid Deep Field South (EDFS)
3. Created a geometric mask to select only the region of EDFS that overlaps DP1 (Euclid's EDFS is $\sim 20 \text{ deg}^2$)
4. Spatial match of Euclid objects to DP1 objects -> merged catalog



Star-Galaxy Separators

According to Euclid documentation: “Notice that this classifier (`POINT_LIKE_PROB`) is heavily biased towards a high purity, and thus has a low completeness.”

`MUMAX_MINUS_MAG` = The difference between `MU_MAX` (Peak surface brightness above the background in the detection band (directly from `SExtractor`)) and `MAG_STARGAL_SEP` (Magnitude used to compute `POINT_LIKE_PROB`)

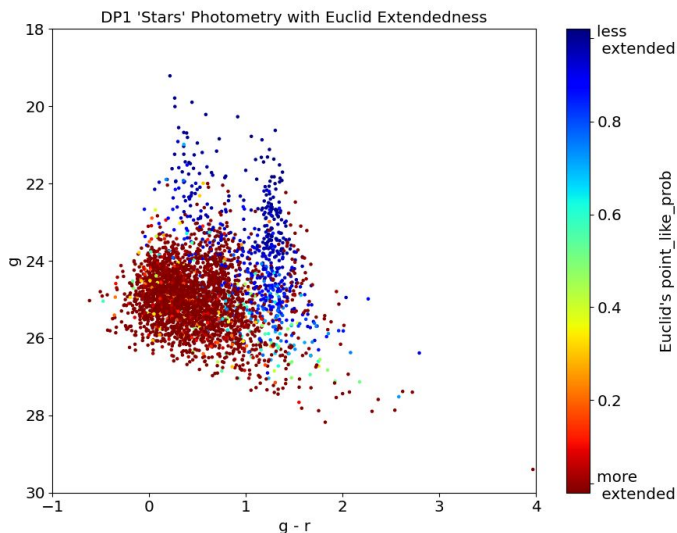
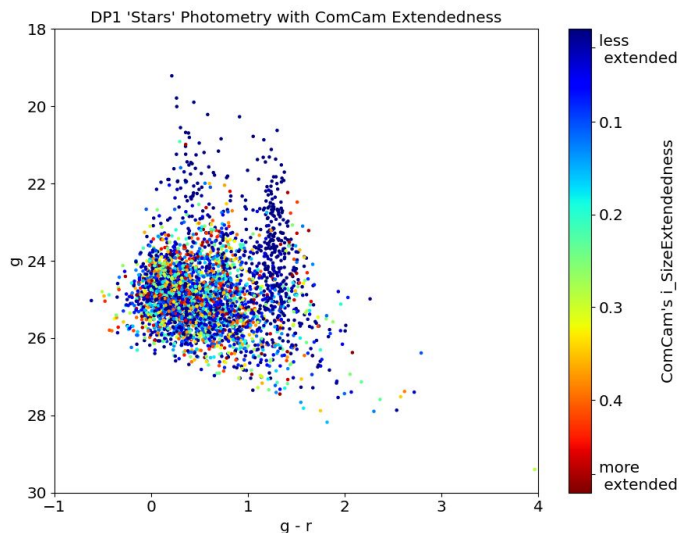


Color-Magnitude of ComCam Stars

For ComCam's stellar sample (selected with `'i_SizeExtendedness' < 0.5`)

-> Euclid's `'POINT_LIKE_PROB'` shows distinct star and galaxy loci in color-magnitude space better than ComCam's `'i_SizeExtendedness'`, especially for faint objects ($g > 24$)

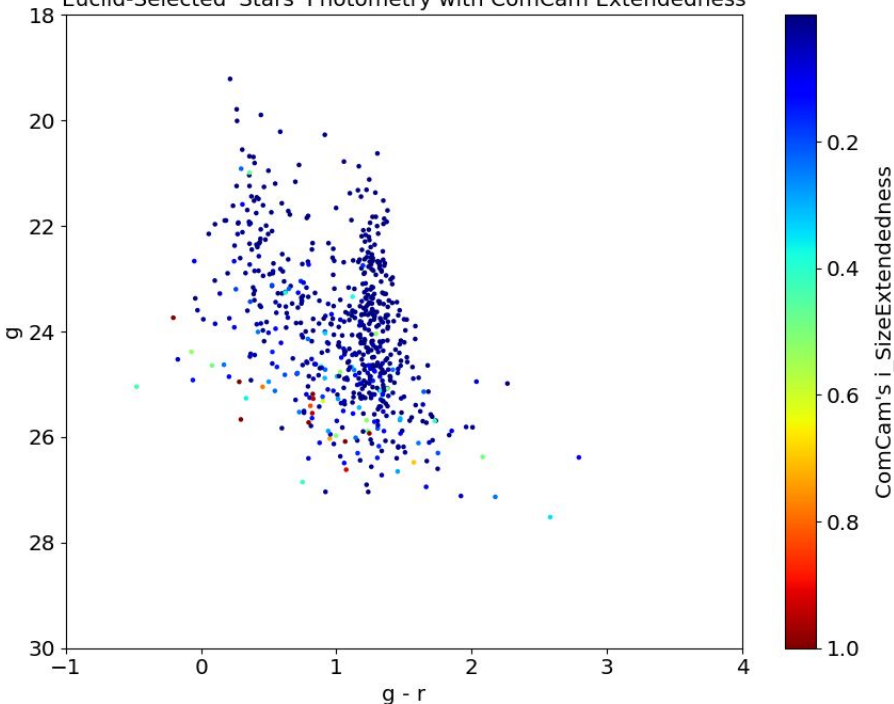
-> Euclid shows that a large fraction of the ComCam-selected "stellar sample" is actually misclassified galaxies, especially for $g > 24$ & $g - r < 1$



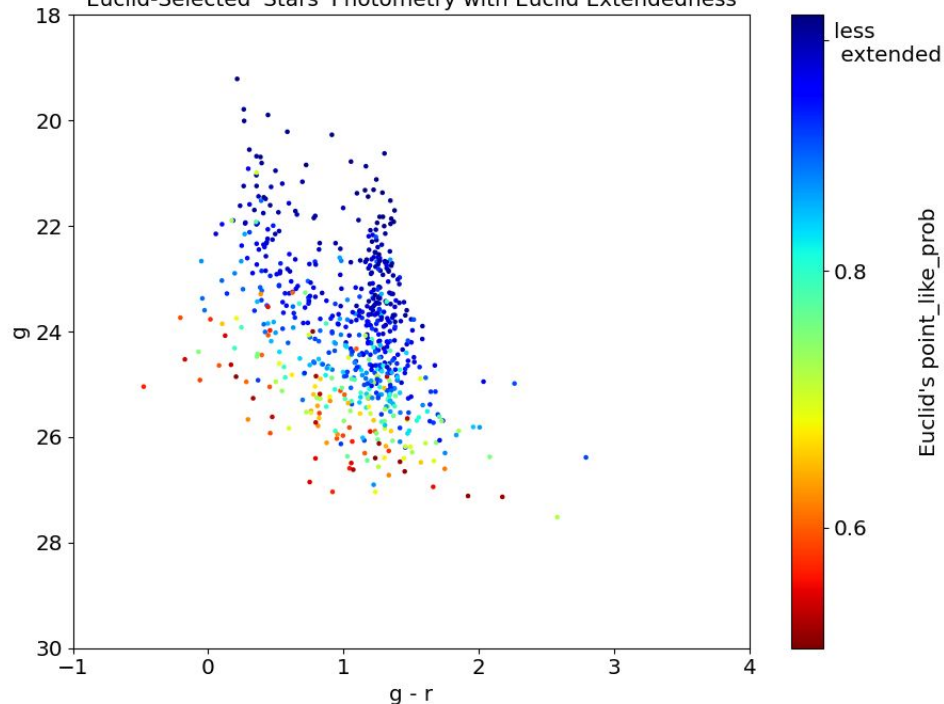
Color-Magnitude of Euclid Stars

Same process as previous slide but now using Euclid selection ('POINT_LIKE_PROB' > 0.5)

Euclid-Selected 'Stars' Photometry with ComCam Extendedness



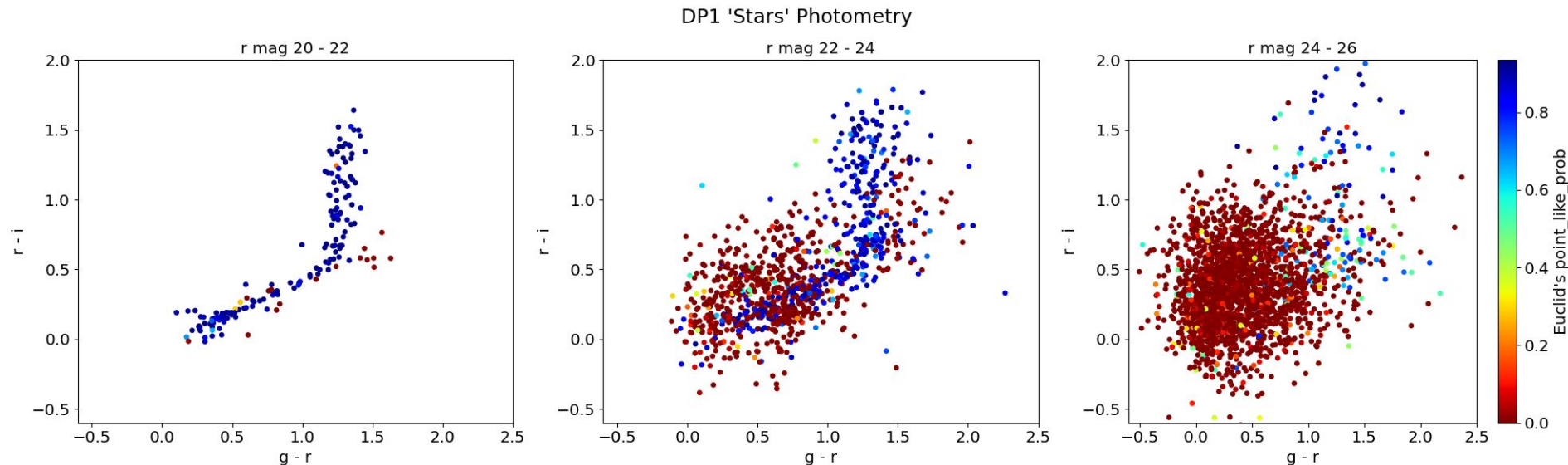
Euclid-Selected 'Stars' Photometry with Euclid Extendedness



Color-Color at different magnitudes

Again, ComCam-selected stellar sample ($i_SizeExtendedness < 0.5$)

-> Based on Euclid's extendedness classification, the bright magnitudes are good at having only stars, 24-26 has mostly galaxies



Summary and Next Steps

- It seems (at least in the short-term) Euclid's `POINT_LIKE_PROB` has higher purity than ComCam's `{band}_sizeExtendedness`
- Applying code to analyze data from LSSTCam observations of EDFs and ECDFs (currently only available to members of Project team)
- Developing a stellar selection that optimally combines data from Euclid and Rubin
 - Using classifications from Euclid to improve and more fully characterize the stellar selection from Rubin
 - Perhaps Rubin is more complete but Euclid more pure
- Analysis notebook: https://github.com/keexcell/star-gal_sep
 - Cleaning this up with documentation for those interested
 - Currently running on USDF. Could get this working on NERSC if we upload Euclid data to NERSC too

Extra plot (more of a question)

Based on how I understand Comcam's star-galaxy sep (i band psf mag - cmodel fit mag), this is the Euclid equivalent (vis band psf - sersic fit mag).

However, what I showed on the previous slide is what their documentation pointed to for S-G sep

